

Genetic Conditions: Frame a Safe Research Question

I can turn the SoW topic into a respectful research question, plan a school-approved source search, and separate biological explanation from personal or medical claims.

[Open the original interactive lesson](#)

Arrival · quiet retrieval

Choose a private statement: I remember gene/allele links; I need a diagram; I need a full recap. No personal history is requested.

Gene and allele A gene is a section of DNA involved in a characteristic. Alleles are different versions of a gene. Retrieve these ideas without naming a real person's genotype.

Genotype and phenotype Genotype describes an allele combination in a model. Phenotype is the observed characteristic. A source may describe patterns, but it cannot diagnose a pupil.

Arrival · quiet retrieval

Probability and certainty An inherited combination can affect probability. Probability is not a promise about one person, and a presentation must not turn a population pattern into an individual prediction.

Research boundary The C44 task is to research a teacher-approved genetic condition and present. It does not ask for family histories, test results, symptoms or medical decisions.

Supported

Choose a teacher-approved question frame and two prepared source cards; identify what each source can answer.

Arrival · quiet retrieval

Standard

Refine a bounded question, plan two-source triangulation and identify one privacy or interpretation risk.

Starter · create the question

Which question is safer and more scientific: one asking how evidence supports an inheritance explanation, or one asking pupils to disclose family information? Explain the boundary.

Picture first: a good question is about the biology. Personal and family disclosure is never part of this task.

Question test

Starter · create the question

A useful research question is answerable from approved sources, uses neutral language, has a clear science focus and leaves diagnosis, advice and personal disclosure outside the task. If the wording assumes its conclusion, rewrite it before searching.

- A. How do approved sources explain the inheritance pattern and its uncertainty?
- B. Who in our class or families may have this condition?
- C. Can the presentation predict exactly what will happen to one person?

Starter · create the question



Four parts of a respectful research plan

Watch the decision trail

Respectful research question checker

Choose a draft question, then judge it. The condition is the one your teacher approved.

Watch the decision trail

- A. How is the condition inherited, and what does that mean for probability in a Punnett square?
- B. Who in our class might have it?
- C. What do school-approved sources say the condition changes in the body, and how is that known?
- D. Is it their parents' fault?
- E. Where do the sources agree and disagree, and what is still uncertain?
- F. respectful, answerable from biology

Watch the decision trail

G. about people, not biology — rewrite

Find the C44 purpose: research a genetic condition and present an evidence-led explanation.

Label the science needed: gene, allele, inheritance pattern, probability, variation and uncertainty at 1BI0 Foundation core.

Map each part of the question to an approved source field; leave a visible Gap where evidence cannot settle an individual outcome.

Broad topic “A genetic condition” is a topic, not yet a workable question. It gives no boundary for evidence or presentation.

Watch the decision trail

Bounded question “How do two approved sources explain the inherited biology, variation and uncertainty?” identifies evidence that can be found and compared.

Presentation purpose The audience needs a clear Foundation explanation and an honest limit, not a medical profile of a person.

Success decision The question is ready only when it is neutral, source-answerable, private and precise enough to organise the final explanation.

Choose, then justify

Select the checks that make a source useful for this question, not merely easy to read.

Authority Who published the item, for what audience, and under what scientific or public-information responsibility?

Currency What date or review date is visible? A date helps locate the item; it does not make every claim correct.

Relevance Which exact part answers the bounded inheritance question? Do not collect an entire page when one defined idea is needed.

Choose, then justify

Boundary Does the source explain biology without inviting self-diagnosis, personal disclosure or unqualified advice?

A school-approved source set narrows access safely. The learner still makes the scientific choice, records why the source is relevant and keeps one uncertainty visible.

Accuracy includes limits

Use neutral language: a person with a condition, unless a teacher-approved source explains a community preference.

Separate inheritance probability from certainty about any individual person.

Keep medical decisions, diagnosis and personal advice outside the task.

Inside C44 Explain inherited biology, compare approved evidence, credit sources, present variation and state uncertainty.

Outside C44 Ask about symptoms, test results, family history, treatment choices or whether a named person has a condition.

Accuracy includes limits

Repair move Replace personal wording with “What do approved sources explain?” and replace certainty with an evidence-bounded probability statement.

Escalation If a pupil raises a personal concern, pause the public task and use the school's approved safeguarding or health-support route.

Build the route together

1 · Locate

Write a bounded question and identify two teacher-approved source cards or locally saved extracts.

2 · Connect

Match each source type to one science field: inherited biology, probability, variation, evidence or uncertainty.

3 · Limit

Build the route together

Write one statement the sources must not be used to make about an individual person.

Keywords Use the approved condition name plus inheritance, gene or allele, evidence, probability and variation. Avoid a real person's name or family detail.

Source record Prepare fields for publisher, title, date or review date, purpose, local locator and the exact idea used.

Two-source reason One source can be incomplete. A second source allows agreement, emphasis and uncertainty to be compared rather than copied.

Build the route together

Presentation route Decide whether the final explanation will be a private talk, four-panel card, narrated local slide or another agreed accessible form.

Current-main LAUNCH Science W12 DNA and W13 inheritance lessons, pack curriculum alignment, and teacher-approved local source cards; the HTML opens no live URL.

Rehearse the plan aloud or by pointing. The audience checks whether every field serves C44 and whether privacy, respect and evidence limits are explicit before the independent brief begins.

Make the evidence trail usable

Create a research brief for one teacher-approved genetic condition using no personal or family data. Record the question, keywords, two approved source types, the evidence categories required, a respectful-language check and one uncertainty that must remain visible.

A · Question Write one neutral, source-answerable question. Underline the science concept and circle the word that keeps the claim bounded.

B · Search map List two approved source types and the field each must answer. Add publisher, date, purpose and local locator boxes.

Make the evidence trail usable

C · Evidence categories Prepare headings for inherited biology, source evidence, probability, variation, lived-experience variation and uncertainty.

D · Privacy test Check that the plan requests no pupil or family history, diagnosis, symptoms, test results, treatment choice or medical advice.

Supported

Choose a teacher-approved question frame and two prepared source cards; identify what each source can answer.

Standard

Make the evidence trail usable

Refine a bounded question, plan two-source triangulation and identify one privacy or interpretation risk.

Stretch

Compare how question wording could bias the search, then design a neutral alternative and an uncertainty check.

One evidence link, one limit, one next step

Submit the question and source plan privately. State one claim the research may support and one claim it must not make.

Evidence link Complete: “My question asks ... The first approved source should help because ...”

Honest limit Complete: “The research cannot decide ... about one individual person.”

Next action Name the first source field to complete in W14L2 and the locator that will keep it checkable.

One evidence link, one limit, one next step

Audience response The adult names back the question and privacy boundary, then gives a realistic report-back point.